Lecture plan · for the lecturer

Agents, environments and decisions

AI agents · University · Lecture 1

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Specify agents with PEAS.
- Model decisions as MDPs.
- Design rewards carefully.

Activity: Break the reward

1. Warm-up (5 min): Show the cleaning agent's loop and ask students where the bug is.
2. Specify (15 min): Pairs write PEAS and a reward for a gridworld exit task.
3. Hack (15 min): Swap rewards and find a policy that scores high without solving the task.
4. Repair (20 min): Fix the reward and compute two steps of value iteration by hand.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which loophole surprised you?
- What does γ change?
- How will you assess reward design?

Exit ticket

- Write a reward for a task you know and one way it could be gamed.

Try it in your next lecture

Give pairs a paper gridworld and ask them to write a reward for reaching the exit; swap rewards between pairs and let each find a loophole in the other’s design.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **Which reward best stops the drop-and-collect loop?**

◯ Reward how clean the room is ◯ No reward at all ◯ More reward per pickup

2 **Compute returns.**

With γ = 0.9 and rewards 0, 0, 10, the return from the start is _____. With γ = 0.5 it is _____.

Word bank: 10 · 8.1 · 2.5 · 5

3 **A campus delivery robot. Which are correct PEAS entries? Choose all that apply.**

☐ Sensors: the delivery schedule's owner ☐ Environment: paths, people, weather

☐ Performance: on-time, undamaged deliveries ☐ Actuators: wheels and a locker door

4 **If the reward function is implemented without bugs, the agent will pursue the designer's intended goal.**

◯ True ◯ False

5 **Match each environment property to an example.**

Partially observable	• • • •	Driving in traffic
Dynamic	• • • •	Poker
Episodic	• • • •	Chess
Fully observable and deterministic	• • • •	Classifying one image

Answers · for the lecturer

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1 **Reward how clean the room is**

Reward the outcome, not the action.

2 **With γ = 0.9 and rewards 0, 0, 10, the return from the start is 8.1. With γ = 0.5 it is 2.5.**

$0.9^2 \times 10 = 8.1$ and $0.5^2 \times 10 = 2.5$: a smaller γ makes distant rewards count less.

3 **Performance: on-time, undamaged deliveries; Environment: paths, people, weather; Actuators: wheels and a locker door**

Sensors are cameras, lidar and GPS, not people.

4 **False**

Bug-free code can still specify the wrong objective; that is specification gaming.

5 **Partially observable — Poker**
Dynamic — Driving in traffic
Episodic — Classifying one image
Fully observable and deterministic — Chess

Poker hides the other players' cards (partially observable); traffic keeps changing while the driver decides (dynamic); each image is classified on its own (episodic); a chess board is fully visible and every move has a certain result (fully observable and deterministic).

For you to think about

Where in your course do students design a reward or objective themselves, and do they ever see it fail?